

NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 2 EXAMINATION 2014-2015

EE7403 – IMAGE ANALYSIS & PATTERN RECOGNITION

April / May 2015

Time Allowed: 3 hours

INSTRUCTIONS

1. This paper contains 5 questions and comprises 3 pages.
 2. Answer ALL questions.
 3. All questions carry equal marks.
 4. This is a closed-book examination.
 5. Unless specifically stated, all symbols have their usual meanings.
-

1. A digital image filter impulse response is given by

$$h(x, y) = \delta(x-1, y-1) + \delta(x-1, y) + \delta(x, y-1) \\ - \delta(x+1, y) - \delta(x, y+1) - \delta(x+1, y+1).$$

- (a) Show the filter mask and indicate your x - and y -axes besides the mask.
(6 Marks)
- (b) Express the output image $g(x, y)$ in terms of the input image $f(x, y)$.
(6 Marks)
- (c) Compute the Fourier transform of the impulse response of the filter $H(u, v)$ given by:

$$H(u, v) = \sum_{x=-\infty}^{\infty} \sum_{y=-\infty}^{\infty} h(x, y) \exp[-j2\pi(ux + vy)]$$

(8 Marks)

EE7403

2. During image acquisition, the camera undergoes uniform planar motion in the horizontal direction (x -direction) for a time T_1 and then switches to the vertical direction (y -direction) for a time T_2 . The duration of the camera exposure is $T = T_1 + T_2$. Let the time varying components of motion in x - and y -directions are $x_o(t) = 3t / T_1$ and $y_o(t) = 4t / T_2$, respectively.

(a) Express the acquired image $g(x, y)$ blurred by motion as a function of the image $f(x, y)$ that has no motion blurring.

(10 Marks)

(b) Derive an expression for the blurring function $H(u, v)$.

(10 Marks)

3. (a) An algorithm has detected n points from an image and it is required to detect lines from the image based on the detected points by applying Hough transform. Assume that there is no error caused by the discrete accumulator cells in the parameter space. What are the possible numbers of peaks in the accumulator cells of the parameter space if $n=5$? For each possible number of peaks, give the value of the each peak and state what kinds of lines (the number of pixels on the line) are detected. (Note that a peak means a point whose value is larger than its eight-neighbors.)

(10 Marks)

(b) Four image gradients are represented by complex numbers as $s_1 = a + ja$, $s_2 = b$, $s_3 = -a - ja$, $s_4 = -jb$. Let $c = s_1 + s_2 + s_3 + s_4$ and $d = s_1^2 + s_2^2 + s_3^2 + s_4^2$.

i. Compute the angle of c and compare it with the angles of s_1, s_2, s_3 and s_4 .

ii. Compute the angle of d and compare it with the angles of s_1, s_2, s_3 and s_4 .

(10 Marks)

4. In a school, the height of male students follows a Gaussian PDF and the height of the female students also follows a Gaussian PDF. A general high-dimensional Gaussian PDF is given below:

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma_i|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_i)^T \Sigma_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) \right].$$

Note: Question No. 4 continues on page 3

EE7403

You are in an office in the school campus. A student of height 1.67m knocks at the door. You need to judge whether he or she is a male or female student with the minimal probability of wrong decision.

- (a) What are your decision and the probability that your decision is wrong if you know that there are 300 male students and 700 female students in the school?

(5 Marks)

- (b) What is your decision if you know that there are equal number of male and female students and the average height of male students is 1.7m and that of female students is 1.62m ?

(7 Marks)

- (c) What is your decision if you have the information in (b) and further know that the variance of the heights of male students is 0.04 and that of female students is 0.09 ?

(8 Marks)



Consider a database of c classes, where each class has n samples and each sample has m features. Each element of the database is represented by x_{kij} , $1 \leq i \leq m$, $1 \leq j \leq n$, $1 \leq k \leq c$.

- (a) Compute the class-conditional mean μ_{ki} and the element σ_{kpq} of the covariance matrix.

(5 Marks)

- (b) Construct column feature vector $\mathbf{x}_{kj}$ of each sample for given x_{kij} and compute the class-conditional mean vector $\boldsymbol{\mu}_k$ and the covariance matrix $\boldsymbol{\Sigma}_k$ using $\mathbf{x}_{kj}$.

(5 Marks)

- (c) Construct the class-conditional data matrix $\mathbf{X}_k$ using $\mathbf{x}_{kj}$ and $\boldsymbol{\mu}_k$ such that the mean feature vector of $\mathbf{X}_k$ is zero, and compute $\boldsymbol{\Sigma}_k$ using $\mathbf{X}_k$.

(5 Marks)

- (d) It is necessary to extract a one-dimensional feature. Let the unit length column vector that span this dimension in the feature space is $\mathbf{a}$. Derive the class-conditional variance v_k of the one-dimensional feature from the definition of the variance using $\mathbf{a}$ and $\mathbf{X}_k$ and express the final result in terms of $\mathbf{a}$ and $\boldsymbol{\Sigma}_k$.

(5 Marks)

END OF PAPER

4.

In a school, the height of male students follows a Gaussian PDF and the height of the female students also follows a Gaussian PDF. A general high-dimensional Gaussian PDF is given below:

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\Sigma_i|^{1/2}} \exp \left[-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu}_i)^T \Sigma_i^{-1} (\mathbf{x} - \boldsymbol{\mu}_i) \right].$$

You are in an office in the school campus. A student of height 1.67m knocks at the door. You need to judge whether he or she is a male or female student with the minimal probability of wrong decision.

- (a) What are your decision and the probability that your decision is wrong if you know that there are 300 male students and 700 female students in the school?

(5 Marks)

$$p(w_1) = 0.3 \quad p(w_2) = 0.7$$

$$p_1(x) = \frac{1}{(2\pi)^{d/2} |\Sigma_1|^{1/2}} \exp \left[-\frac{1}{2} (x - \mu_1)^T \Sigma_1^{-1} (x - \mu_1) \right]$$

$$p_2(x) = \frac{1}{(2\pi)^{d/2} |\Sigma_2|^{1/2}} \exp \left[-\frac{1}{2} (x - \mu_2)^T \Sigma_2^{-1} (x - \mu_2) \right]$$

5.

Consider a database of c classes, where each class has n samples and each sample has m features. Each element of the database is represented by x_{kij} , $1 \leq i \leq m$, $1 \leq j \leq n$, $1 \leq k \leq c$.

- (a) Compute the class-conditional mean μ_{ki} and the element σ_{kpq} of the covariance matrix.

(5 Marks)

- (b) Construct column feature vector $\mathbf{x}_{kj}$ of each sample for given x_{kij} and compute the class-conditional mean vector $\boldsymbol{\mu}_k$ and the covariance matrix $\boldsymbol{\Sigma}_k$ using $\mathbf{x}_{kj}$.

(5 Marks)

(a). c class n samples each class m features.

$$\mu_{ki} = \frac{1}{n} \sum_{j=1}^n x_{kij}$$

$$\sigma_{kpq} = (\mathbf{x}_{kp} - \mathbf{x}_{kq}) (\mathbf{x}_{kp} - \mathbf{x}_{kq})^T$$

$$(b) \quad \mathbf{x}_{kj} = [x_{k1j} \quad x_{k2j} \quad \dots \quad x_{kmj}]^T \quad \boldsymbol{\mu}_k = [\mu_{k1} \quad \mu_{k2} \quad \dots \quad \mu_{km}]^T$$

$$\boldsymbol{\Sigma}_k = \frac{1}{n} \sum_{j=1}^n (\mathbf{x}_{kj} - \boldsymbol{\mu}_k) (\mathbf{x}_{kj} - \boldsymbol{\mu}_k)^T$$

- (c) Construct the class-conditional data matrix $\mathbf{X}_k$ using $\mathbf{x}_{kj}$ and $\boldsymbol{\mu}_k$ such that the mean feature vector of $\mathbf{X}_k$ is zero, and compute $\boldsymbol{\Sigma}_k$ using $\mathbf{X}_k$.

(5 Marks)

Let $\tilde{\mathbf{x}}_k = [x_{k1} - \mu_{k1}, x_{k2} - \mu_{k2}, \dots, x_{kj} - \mu_{kj}, \dots, x_{kn} - \mu_{kn}]$ $\Rightarrow$ the mean feature vector is 0

$$\begin{aligned} \boldsymbol{\Sigma}_k &= \frac{1}{n} \sum_{j=1}^n (\tilde{\mathbf{x}}_k - \mathbf{r}_k) (\tilde{\mathbf{x}}_k - \mathbf{r}_k)^T \\ &= \frac{1}{n} \sum_{j=1}^n \tilde{\mathbf{x}}_k \tilde{\mathbf{x}}_k^T \end{aligned}$$

- (d) It is necessary to extract a one-dimensional feature. Let the unit length column vector that span this dimension in the feature space is $\mathbf{a}$. Derive the class-conditional variance v_k of the one-dimensional feature from the definition of the variance using $\mathbf{a}$ and $\mathbf{X}_k$ and express the final result in terms of $\mathbf{a}$ and Σ_k .

(5 Marks)

END OF PAPER

$$a_{m \times 1} \quad a^T_{1 \times m} \quad a^T a = 1$$

$$\alpha_i = a^T \tilde{x}_{ki} \quad \alpha = [\alpha_1, \alpha_2, \dots, \alpha_n] = a^T [\tilde{x}_{k1}, \tilde{x}_{k2}, \dots, \tilde{x}_{kn}]$$

$$= a^T x_k$$

$$v_k = \frac{1}{n} \alpha \alpha^T = \frac{1}{n} a^T x_k (a^T x_k)^T$$

$$= \frac{1}{n} a^T x_k x_k^T a$$

$$= a^T \Sigma_k a$$